LINEAR ALGEBRA

ASSIGNMENT 2.

DETERMINENTS.

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What is a matrix Determinent?

Determinent is the scalar property of a square matrix. It is a unique number associated to the matrix.

We can find the determinant of a matrix by the following method.

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow |A| = ad - bc \text{ — for } 2\times2.$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Rightarrow a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$\Rightarrow a(ei - fh) - b(di - fg) + c(dh - eg) \text{ — for } 3\times3.$$

Determinants can be used to calculate areas.

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Properties of Matrix Determinants.

Following are the properties of determinants.

1) The determinant of a matrix and its transpose is equal. $|A| = |A^t|$

2) If a matrix has two identical rows/columns, then determinant is zero. $|A| = 0$.

3) If all entries of a row/column is zero, then determinant is zero. $|A| = 0$

4) If two rows/colums are interchanged, then $|A| = -|A|$.

5) If entries of a row/column are multiplied by a number $k \in \mathbb{R}$, then resulting determinant is $k|A|$, $k \neq 0$.

6) If each entry of row/column consists of two terms, then its determinant can be written as sum of the two determinants.

7) If to each entry of a row/column of a matrix A is added a non zero multiple of corresponding entry of another row or column, then determinant is $|A|$

8) If a matrix is in triangular form, then the value of its determinant is the product of the entries on its main diagonal.

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Properties of Determinents with examples.

1) For a square matrix. The determinent of the matrix and it's transpose is the same. $|A| = |A^t|$

If A is a matrix, then

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \Rightarrow \begin{vmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{vmatrix} = |A^t|$$

$$\Rightarrow a_{11}a_{22} - a_{12}a_{21} = a_{11}a_{22} - a_{21}a_{12}$$

Example,

let $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, then $|A| = 2(2) - 3(1) \Rightarrow 1.$

$$|A^t| = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(3) = 1.$$

2) If a square matrix has two identical rows or columns, then $|A| = 0$.

If A is a matrix, then

$$|A| = \begin{vmatrix} a_{11} & 2 & 1 \\ a_{21} & 2 & 1 \\ a_{31} & 2 & 1 \end{vmatrix} \Rightarrow a_{11}(2-2) - a_{21}(2-2) + a_{31}(2-2) = 0.$$

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Example:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 5 & 2 & 1 \\ 7 & 2 & 1 \end{bmatrix}$$

$|A| = 3(2-2) - 5(2-2) + 7(2-2) \Rightarrow 0.$

3) If all the entries of a row or column in a matrix is 0, then determinent is zero.

If a A is a matrix, then

$$A = \begin{bmatrix} a_{11} & 0 \\ a_{21} & 0 \end{bmatrix} \Rightarrow |A| = a_{11}(0) - a_{21}(0) = 0.$$

Example.-

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}, \quad |A| = 0(2) - 0(1) = 0.$$

4) If two rows or columns are interchanged, the resulting determinent is $-|A|$.

If A is a matrix, then

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = - \begin{vmatrix} a_{12} & a_{11} \\ a_{22} & a_{21} \end{vmatrix} = -|A|$$

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Example:

$$A = \begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix} \Rightarrow |A| = 5(2) - 3(1) = 7.$$

$$|A| = -\begin{vmatrix} 1 & 5 \\ 2 & 3 \end{vmatrix} = -[1(3) - 5(2)] = -(-7) = 7.$$

5) If the entries of a row or column are multiplied by a number $k \in \mathbb{R}$, then resulting determinent is $k|A|$.

let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, then

$$k.|A| = \begin{vmatrix} a_{11} & ka_{12} \\ a_{21} & ka_{22} \end{vmatrix} \Rightarrow$$

Example:

$$A = \begin{bmatrix} 7 & 2 \\ 4 & 6 \end{bmatrix}, \quad k = 3$$

$$|A| = 7(6) - 4(2) = 34 \Rightarrow k|A| = 3(34) \Rightarrow 102.$$

$$k|A| = \begin{vmatrix} 7 & 3(2) \\ 4 & 3(6) \end{vmatrix} = 7(18) - 4(6) = 126 - 24 = 102.$$

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6) If each entry of a row or column consists of two terms, then it's determinent can be written as the sum of two determinents.

Example.

$$A = \begin{bmatrix} a_{11}+b_{11} & a_{12} & a_{13} \\ a_{21}+b_{21} & a_{22} & a_{23} \\ a_{31}+b_{31} & a_{32} & a_{33} \end{bmatrix}, \text{ then}$$

$$\begin{vmatrix} a_{11}+b_{11} & a_{12} & a_{13} \\ a_{21}+b_{21} & a_{22} & a_{23} \\ a_{31}+b_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} b_{11} & a_{12} & a_{13} \\ b_{21} & a_{22} & a_{23} \\ b_{31} & a_{32} & a_{33} \end{vmatrix}$$

7) If to each entry of a row or column of a matrix A is added a non-zero multiple of corresponding entry of another row/column, then the determinent is $|A|$

Example.

$$\text{let } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$\text{if } |B| = \begin{vmatrix} a_{11} & a_{12}+ka_{11} \\ a_{21} & a_{22}+ka_{21} \end{vmatrix} \Rightarrow \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & ka_{11} \\ a_{21} & ka_{21} \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + 0 \Rightarrow \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = |A|$$

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8) If a matrix is in triangular form, then the value of its determinent is the product of the entries on it's main diagonal.

Example:

let $A = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow 1 \times 2 \times 3 = 6. = |A|$

$|A| = 1(6-0) - 5(0-0) + 7(0-0) = 6.$

let $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \Rightarrow |B| = 24.$

$|B| = 2(12-0) + 0(0-0) + 0(0-0) = 24.$

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